

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2013-14

Botany

Paper: BOB-602 — Microbiology and Plant Pathology

Time: Three hours

Full Marks: 70

Note: Attempt five questions in all, selecting two questions each from Section-A and B. Question No. 1 is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following in not more than 50 words each or fill in the blanks appropriately: [1 × 10 = 10]

- (a) Define lag phase.
- (b) Name two denitrifying bacteria.
- (c) Name two lactic acid bacteria.
- (d) Define T-DNA.
- (e) Name a methanogenic bacterium.
- (f) Name the pathogen of powdery mildew of Sheesham.
- (g) Name two famines that were caused by plant pathogens.
- (h) Define quarantine.
- (i) Name a PR-protein.
- (j) PUSA associated with IARI stands for _____.

SECTION – A

Question 2. Give a detailed account of acetic acid production by different methods. [15]

Question 3. Discuss critically the role of microbes in genetic engineering. [15]

Question 4. Write explanatory notes on any two of the following: [7.5 × 2 = 15]

- (a) A general account on Mycoplasma and Thermoplasma
- (b) Retroviruses
- (c) Mechanism of gene transfer in transduction

Question 5. Write short notes on any three of the following: [5 × 3 = 15]

- (a) Bacterial growth curve

- (b) Scope of Microbiology
- (c) N₂ fixing cyanobacteria
- (d) Nitrification

SECTION – B

Question 6. How host physiology changes following attack by pathogens? Discuss the mechanisms involved with suitable examples. [15]

Question 7. Give an account on Integrated Disease Management (IDM). Why is it considered as the best disease control strategy? [15]

Question 8. Write explanatory notes on any two of the following: [7.5 × 2 = 15]

- (a) Brief history of plant pathology and its scope
- (b) Rust of linseed
- (c) Wilt of tomato

Question 9. Write notes on any three of the following: [5 × 3 = 15]

- (a) Transmission and spread of plant diseases
- (b) Smut of Bajra
- (c) Mosaic of sugarcane
- (d) Haustoria

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